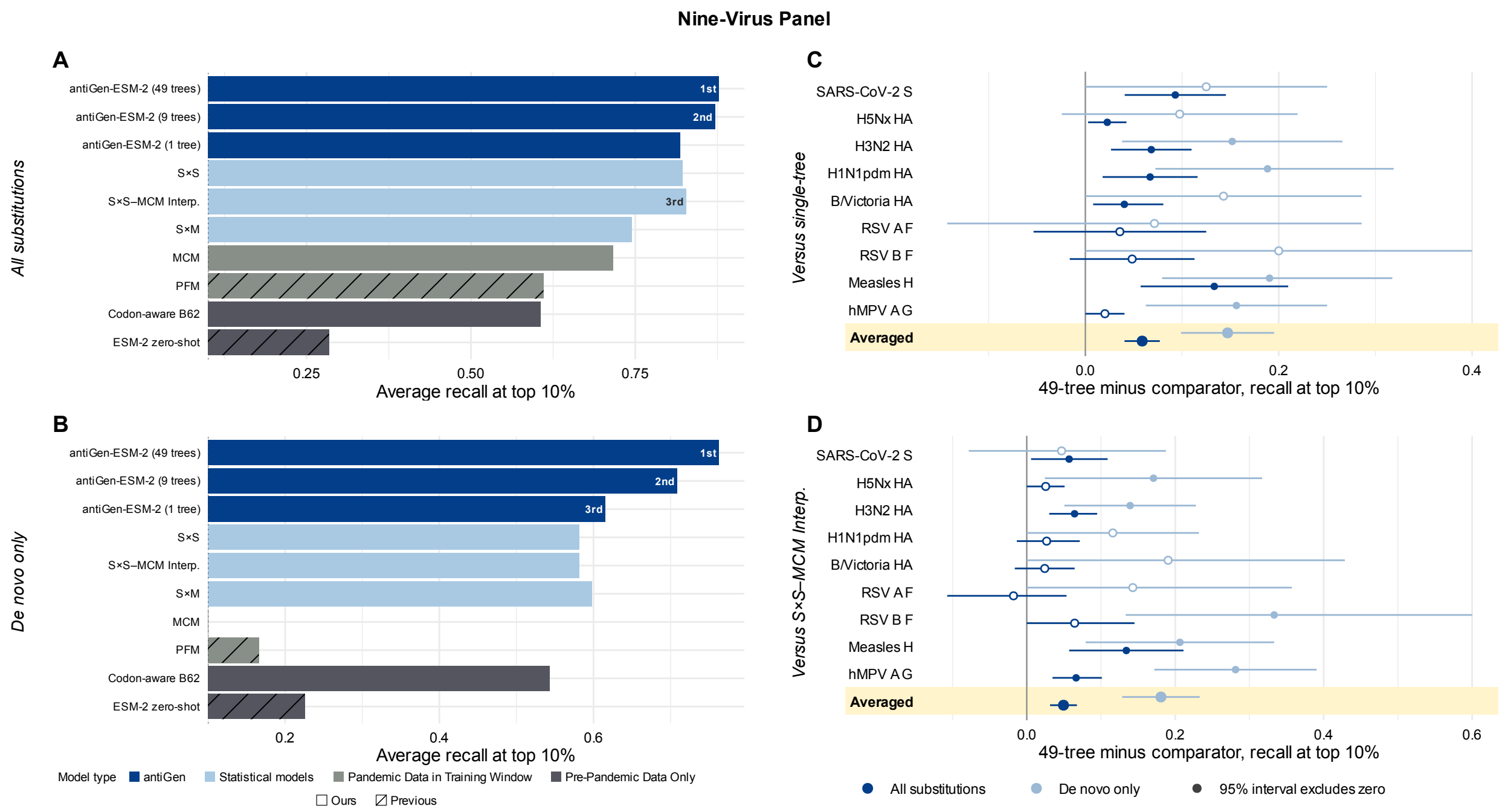
Nine-Virus Panel
A
All substitutions
antiGen-ESM-2 (49 trees)
antiGen-ESM-2 (9 trees)
antiGen-ESM-2 (1 tree)
S×S
S×S–MCM Interp.
S×M
MCM
PFM
Codon-aware B62
ESM-2 zero-shot
1st
2nd
3rd
0.25
0.50
0.75
Average recall at top 10%
B
De novo only
antiGen-ESM-2 (49 trees)
antiGen-ESM-2 (9 trees)
antiGen-ESM-2 (1 tree)
S×S
S×S–MCM Interp.
S×M
MCM
PFM
Codon-aware B62
ESM-2 zero-shot
1st
2nd
3rd
0.2
0.4
0.6
Average recall at top 10%
Model type
antiGen
Statistical models
Pandemic Data in Training Window
Pre-Pandemic Data Only
Ours
Previous
C
Versus single-tree
SARS-CoV-2 S
H5Nx HA
H3N2 HA
H1N1pdm HA
B/Victoria HA
RSV A F
RSV B F
Measles H
hMPV A G
Averaged
0.0
0.2
0.4
49-tree minus comparator, recall at top 10%
D
Versus S×S–MCM Interp.
SARS-CoV-2 S
H5Nx HA
H3N2 HA
H1N1pdm HA
B/Victoria HA
RSV A F
RSV B F
Measles H
hMPV A G
Averaged
0.0
0.2
0.4
0.6
49-tree minus comparator, recall at top 10%
All substitutions
De novo only
95% interval excludes zero